

Data Structures and Algorithms for Problem Solving (CS1.304)

Slip Test (3 Sept 2026)

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Conventional Matrix Multiplication

Suppose you want to multiply two matrices **A** and **B**, each of size 2×2 . Their product is denoted by **C**.

$$\mathbf{A} \times \mathbf{B} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \mathbf{C}$$

The matrix **C** is evaluated as follows (using **8** multiplications).

$$c_{11} = a_{11}b_{11} + a_{12}b_{21}$$

$$c_{21} = a_{21}b_{11} + a_{22}b_{21}$$

$$c_{12} = a_{11}b_{12} + a_{12}b_{22}$$

$$c_{22} = a_{21}b_{12} + a_{22}b_{22}$$

Strassen's Matrix Multiplication

However, $C = A \times B$ can also be evaluated using just **7** multiplications, by defining the **7** variables $m_1, m_2, m_3, \dots, m_7$, and each variable uses only one multiplication.

$$m_1 = (a_{11} + a_{22})(b_{11} + b_{22})$$

$$m_2 = (a_{21} + a_{22})b_{11}$$

$$m_3 = a_{11}(b_{12} - b_{22})$$

$$m_4 = a_{22}(b_{21} - b_{11})$$

$$m_5 = (a_{11} + a_{12})b_{22}$$

$$m_6 = (a_{21} - a_{11})(b_{11} + b_{12})$$

$$m_7 = (a_{12} - a_{22})(b_{21} + b_{22})$$

Example

For example,

$$\begin{aligned}c_{12} &= a_{11}b_{12} + a_{12}b_{22} \\&= a_{11}b_{12} - a_{11}b_{22} \\&\quad + a_{11}b_{22} + a_{12}b_{22} \\&= \mathbf{m_3} + \mathbf{m_5}.\end{aligned}$$

$$\mathbf{m_1} = (a_{11} + a_{22})(b_{11} + b_{22})$$

$$\mathbf{m_2} = (a_{21} + a_{22})b_{11}$$

$$\mathbf{m_3} = a_{11}(b_{12} - b_{22})$$

$$\mathbf{m_4} = a_{22}(b_{21} - b_{11})$$

$$\mathbf{m_5} = (a_{11} + a_{12})b_{22}$$

$$\mathbf{m_6} = (a_{21} - a_{11})(b_{11} + b_{12})$$

$$\mathbf{m_7} = (a_{12} - a_{22})(b_{21} + b_{22})$$

Puzzle 1 [3 marks]

Evaluate c_{11}, c_{21}, c_{22} with $\mathbf{m}_1, \mathbf{m}_2, \mathbf{m}_3, \dots, \mathbf{m}_7$, using only plus and minus (or addition and subtraction) operations.

- $c_{11} = a_{11}b_{11} + a_{12}b_{21}$
- $c_{12} = \mathbf{m}_3 + \mathbf{m}_5$
- $c_{21} = a_{21}b_{11} + a_{22}b_{21}$
- $c_{22} = a_{21}b_{12} + a_{22}b_{22}$

$$\mathbf{m}_1 = (a_{11} + a_{22})(b_{11} + b_{22})$$

$$\mathbf{m}_2 = (a_{21} + a_{22})b_{11}$$

$$\mathbf{m}_3 = a_{11}(b_{12} - b_{22})$$

$$\mathbf{m}_4 = a_{22}(b_{21} - b_{11})$$

$$\mathbf{m}_5 = (a_{11} + a_{12})b_{22}$$

$$\mathbf{m}_6 = (a_{21} - a_{11})(b_{11} + b_{12})$$

$$\mathbf{m}_7 = (a_{12} - a_{22})(b_{21} + b_{22})$$

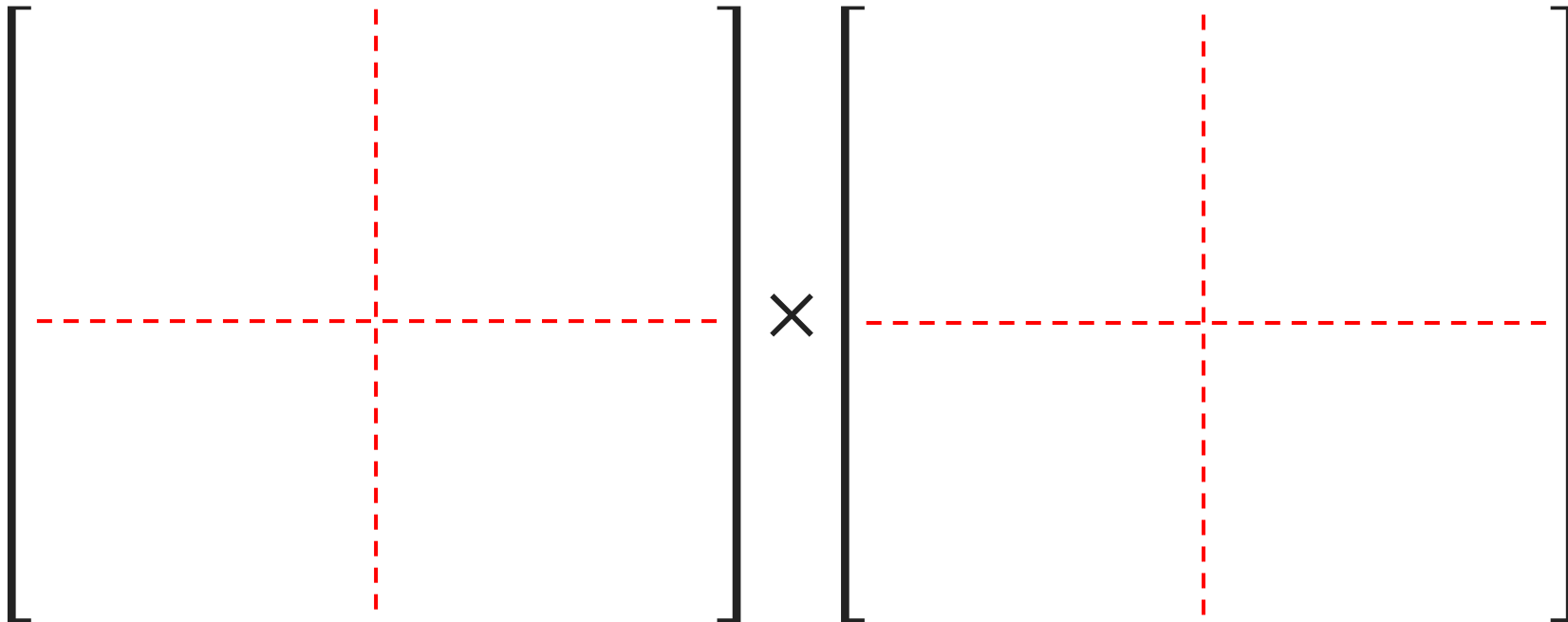
Algorithm (Strassen)

Given two matrices **A** and **B** of size **n** × **n** each, they can both be split into four matrices of size $\frac{n}{2} \times \frac{n}{2}$ each.

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \times \begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nn} \end{bmatrix}$$

Algorithm (Strassen)

Given two matrices ***A*** and ***B*** of size ***n*** × ***n*** each, they can both be split into four matrices of size $\frac{n}{2} \times \frac{n}{2}$ each.



Analysis (Strassen)

Given two matrices A and B of size $n \times n$ each, they can both be split into four matrices of size $\frac{n}{2} \times \frac{n}{2}$ each.

Let $T(n)$ be the number of multiplications required to multiply two $n \times n$ matrices.

$T(1) = 1$, because a 1×1 matrix contains just a single entry.

Puzzle 2 [3 marks]

Given two matrices ***A*** and ***B*** of size ***n*** × ***n*** each, they can both be split into four matrices of size $\frac{n}{2} \times \frac{n}{2}$ each.

Write and explain a recurrence for $T(n)$ in terms of $T\left(\frac{n}{2}\right)$ for Strassen's matrix multiplication. Solve it. (Take **$\log_2 7 \approx 2.81$** .)

BONUS [1 mark]: What is the total number of arithmetic operations ($\times, +, -$) performed by Strassen's algorithm to multiply two matrices ***A*** and ***B*** of size ***n*** × ***n*** each? Explain.

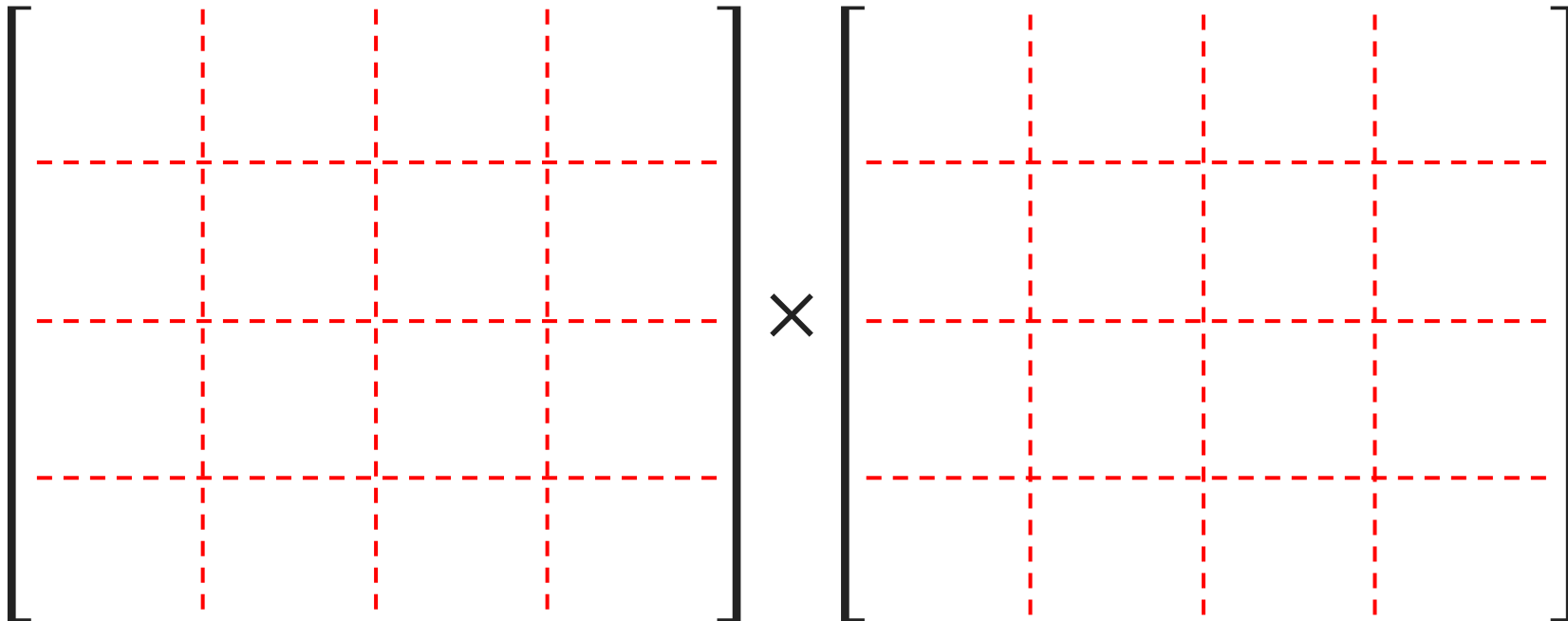
Algorithm (binary matrices)

Given two binary matrices ***A*** and ***B*** of size ***n*** × ***n*** each, they can both be split into sixteen binary matrices of size $\frac{n}{4} \times \frac{n}{4}$ each.

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \times \begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nn} \end{bmatrix}$$

Algorithm (binary matrices)

Given two binary matrices A and B of size $n \times n$ each, they can both be split into sixteen binary matrices of size $\frac{n}{4} \times \frac{n}{4}$ each.



Analysis (binary matrices)

Given two binary matrices A and B of size $n \times n$ each, they can both be split into sixteen binary matrices of size $\frac{n}{4} \times \frac{n}{4}$ each.

Let $T(n)$ be the number of multiplications required to multiply two $n \times n$ matrices.

$T(1) = 1$, because a 1×1 matrix contains just a single entry.

Puzzle 3 [3 marks]

Given two binary matrices A and B of size $n \times n$ each, they can both be split into sixteen binary matrices of size $\frac{n}{4} \times \frac{n}{4}$ each.

In October 2022, Google DeepMind's AI AlphaTensor presented an algorithm that multiplies two 4×4 binary matrices using just 47 multiplications.

Write and explain a recurrence for $T(n)$ in terms of $T\left(\frac{n}{4}\right)$ for binary matrix multiplication. Solve it. (Take $\log_4 47 \approx 2.7773$.)

Thank You!

